

WELCOME TO EEE 4203: Electronics I



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Early development of FET



Drs. Ian Munro Ross (front) and G. C. Dacey jointly developed an experimental procedure for measuring the characteristics of a field-effect transistor in 1955.

Early development of FET



Differences between BJT and FET



The primary difference between the two types of transistors is the fact that:

The BJT transistor is a current-controlled device as depicted in Fig. 6.1a, whereas the JFET transistor is a voltage-controlled device as shown in Fig. 6.1b.

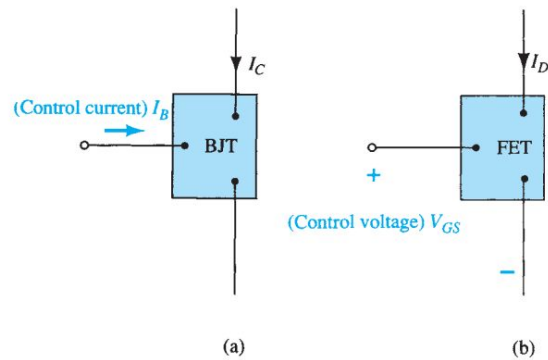


FIG. 6.1

(a) Current-controlled and (b) voltage-controlled amplifiers.

Just as there are *npn* and *pnp* bipolar transistors, there are *n-channel* and *p-channel* field-effect transistors. However, it is important to keep in mind that the BJT transistor is a *bipolar* device—the prefix *bi* indicates that the conduction level is a function of two charge carriers, electrons and holes. The FET is a *unipolar* device depending solely on either electron (*n-channel*) or hole (*p-channel*) conduction.

One of the most important characteristics of the FET is its high input impedance.

At a level of $1\text{ M}\Omega$ to several hundred megohms it far exceeds the typical input resistance levels of the BJT transistor configurations—a very important characteristic in the design of linear ac amplifier systems. On the other hand, the BJT transistor has a much higher sensitivity to changes in the applied signal. In other words, the variation in output current is typically a great deal more for BJTs than for FETs for the same change in the applied voltage.

Differences between BJT and FET



FETs are more temperature stable than BJTs, and FETs are usually smaller than BJTs, making them particularly useful in integrated-circuit (IC) chips.

Three types of FETs are introduced in this chapter: the *junction field-effect transistor* (JFET), the *metal–oxide–semiconductor field-effect transistor* (MOSFET), and the *metal–semiconductor field-effect transistor* (MESFET). The MOSFET category is further broken down into *depletion* and *enhancement* types, which are both described. The MOSFET

Construction of JFET

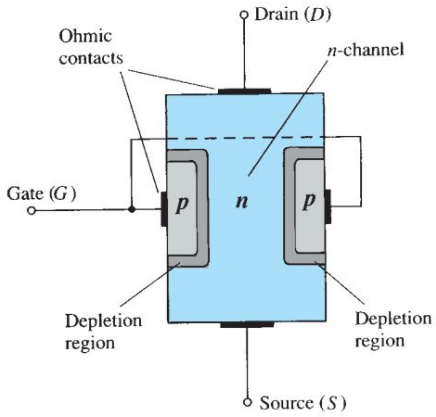
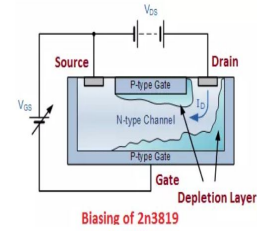


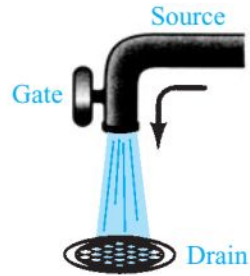
FIG. 6.3

Junction field-effect transistor (JFET).



2N3819 N-Channel JFET





The source of water pressure can be likened to the applied voltage from drain to source, which establishes a flow of water (electrons) from the spigot (source). The “gate,” through an applied signal (potential), controls the flow of water (charge) to the “drain.” The drain and source terminals are at opposite ends of the n -channel as introduced in Fig. 6.3 because the terminology is defined for electron flow.

Symbols

The graphic symbols for the n -channel and p -channel JFETs are provided in Fig. 6.14. Note that the arrow is pointing in for the n -channel device of Fig. 6.14a to represent the direction in which I_G would flow if the p - n junction were forward-biased. For the p -channel device (Fig. 6.14b) the only difference in the symbol is the direction of the arrow in the symbol.

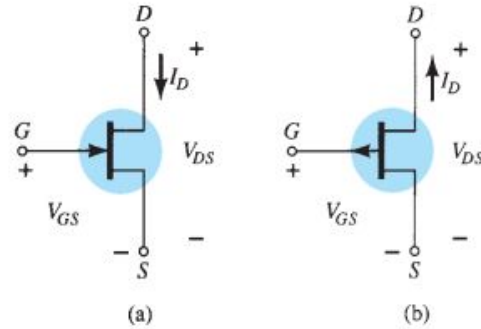


FIG. 6.14

JFET symbols: (a) n -channel; (b) p -channel.

Working Principle of JFET



$V_{GS} = 0 \text{ V}$, V_{DS} Some Positive Value

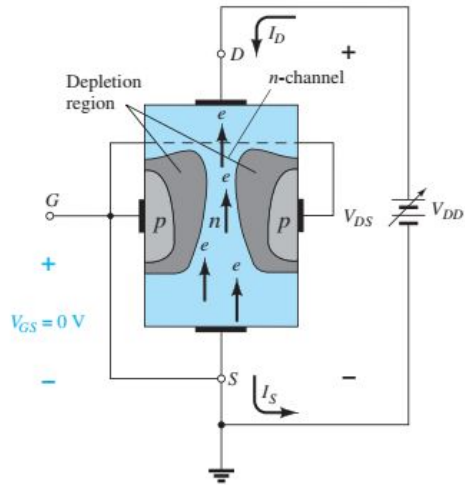


FIG. 6.5

JFET at $V_{GS} = 0 \text{ V}$ and $V_{DS} > 0 \text{ V}$.

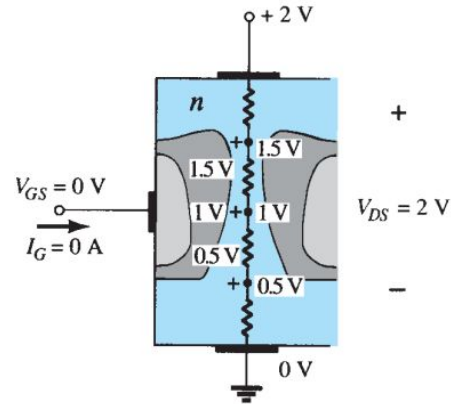


FIG. 6.6

Varying reverse-bias potentials across the p-n junction of an n-channel JFET.

$V_{GS} = 0 \text{ V}$, V_{DS} Some Positive Value

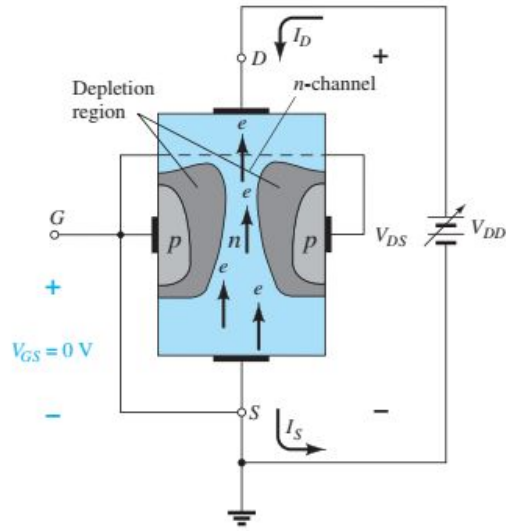


FIG. 6.5

JFET at $V_{GS} = 0 \text{ V}$ and $V_{DS} > 0 \text{ V}$.

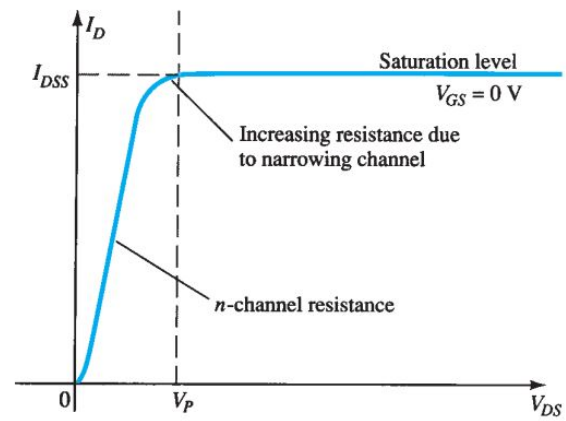


FIG. 6.7

I_D versus V_{DS} for $V_{GS} = 0 \text{ V}$.

$$V_{GS}=0, \quad V_{DS}=V_P$$

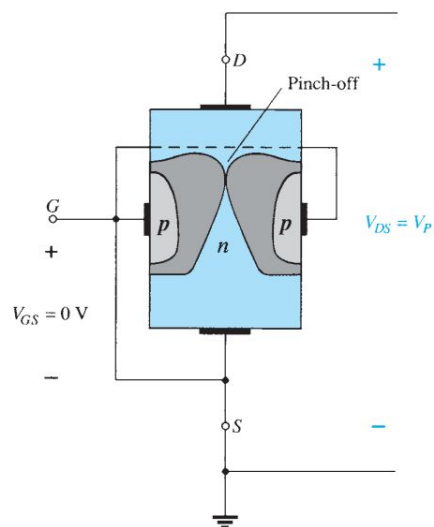


FIG. 6.8
Pinch-off ($V_{GS} = 0 \text{ V}$, $V_{DS} = V_P$).

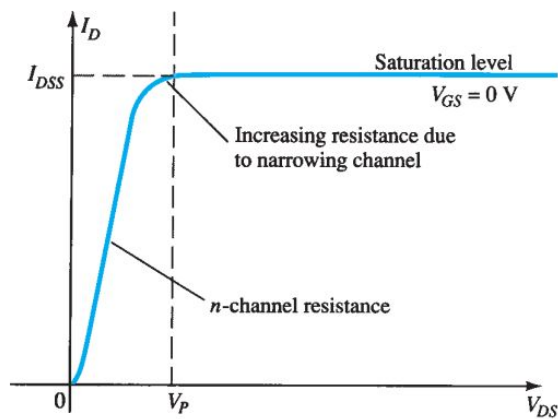


FIG. 6.7
 I_D versus V_{DS} for $V_{GS} = 0 \text{ V}$.

I_{DSS} is the maximum drain current for a JFET and is defined by the conditions $V_{GS} = 0 \text{ V}$ and $V_{DS} > |V_P|$.

$$V_{GS} < 0 \text{ V}$$

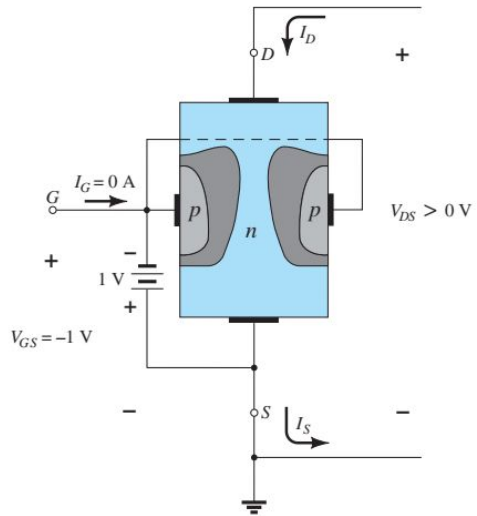


FIG. 6.10

Application of a negative voltage to the gate of a JFET.

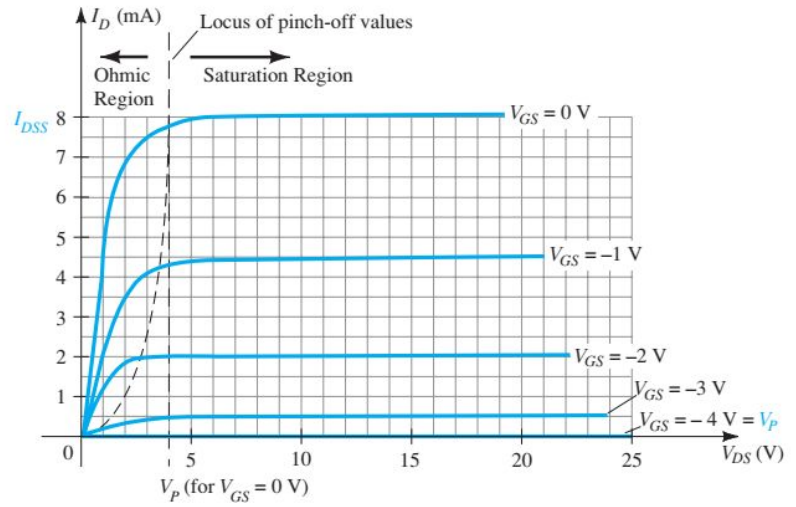


FIG. 6.11

n-Channel JFET characteristics with $I_{DSS} = 8 \text{ mA}$ and $V_P = -4 \text{ V}$.

Voltage-Controlled Resistor

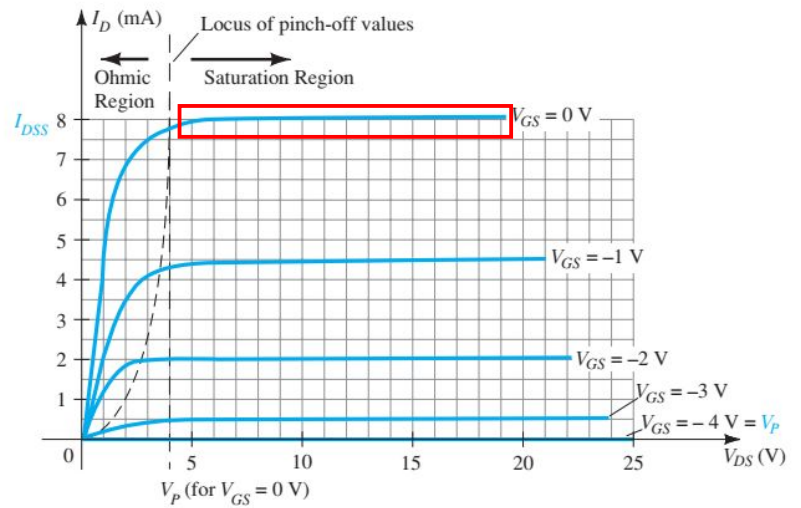
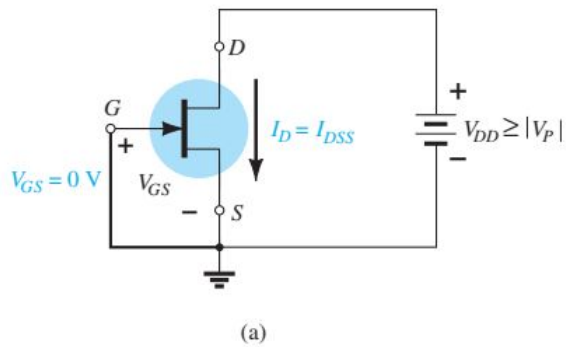
The region to the left of the pinch-off locus of Fig. 6.11 is referred to as the *ohmic* or *voltage-controlled resistance region*. In this region the JFET can actually be employed as a variable resistor (possibly for an automatic gain control system) whose resistance is controlled by the applied gate-to-source voltage. Note in Fig. 6.11 that the slope of each curve and therefore the resistance of the device between drain and source for $V_{DS} < V_P$ are a function of the applied voltage V_{GS} . As V_{GS} becomes more and more negative, the slope of each curve becomes more and more horizontal, corresponding to an increasing resistance level. The following equation provides a good first approximation to the resistance level in terms of the applied voltage V_{GS} :

$$r_d = \frac{r_o}{(1 - V_{GS}/V_P)^2} \quad (6.1)$$

where r_o is the resistance with $V_{GS} = 0$ V and r_d is the resistance at a particular level of V_{GS} .

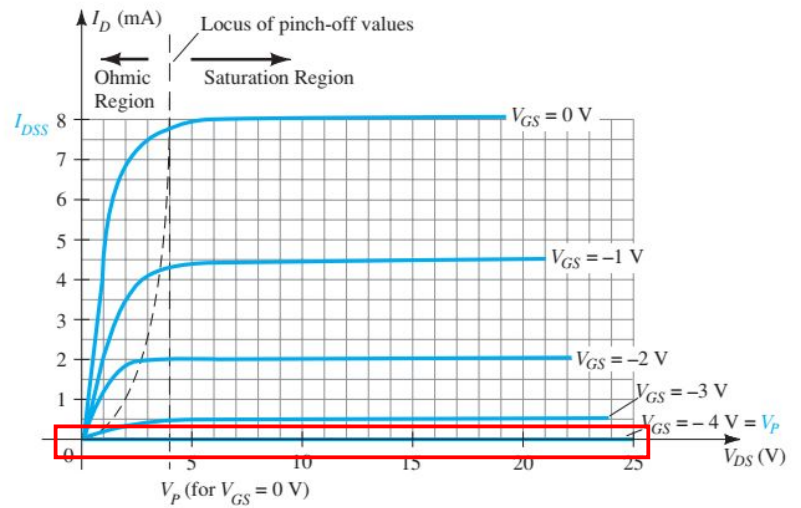
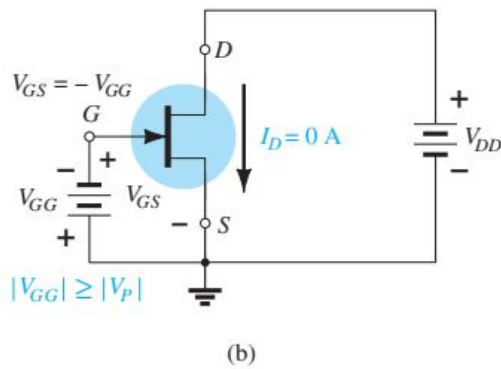
For an n -channel JFET with $r_o = 10$ k Ω ($V_{GS} = 0$ V, $V_P = -6$ V), Eq. (6.1) results in 40 k Ω at $V_{GS} = -3$ V.

Summary

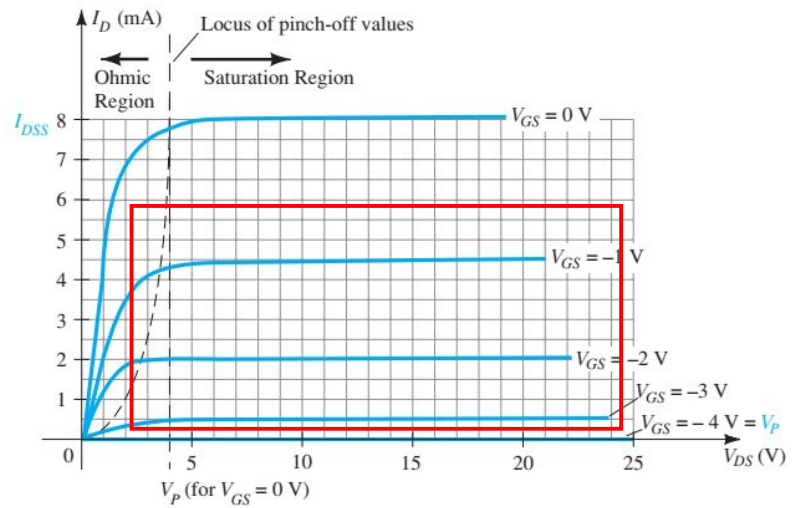
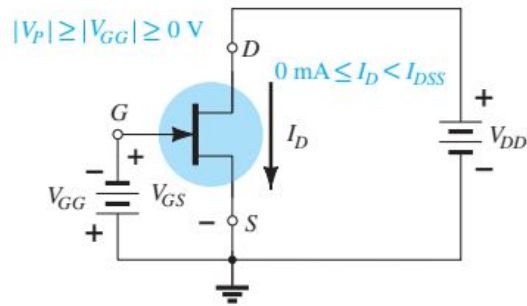


The maximum current is defined as I_{DSS} and occurs when $V_{GS} = 0$ V and $V_{DS} \geq |V_P|$, as shown in Fig. 6.15a.

Summary



For gate-to-source voltages V_{GS} is less than (more negative than) the pinch-off level, the drain current is 0 A ($I_D = 0\text{ A}$), as in Fig. 6.15b.



For all levels of V_{GS} between 0 V and the pinch-off level, the current I_D will range between I_{DSS} and 0 A, respectively, as in Fig. 6.15c.

TASK (5 MARKS)



1. Draw the basic construction of a P-channel JFET.
2. For $V_{GS} = 0V$, briefly describe the working principle of P-channel JFET.
3. Apply proper biasing in the construction and show current flow direction.
4. Sketch the drain characteristics curve for different values of V_{GS} .

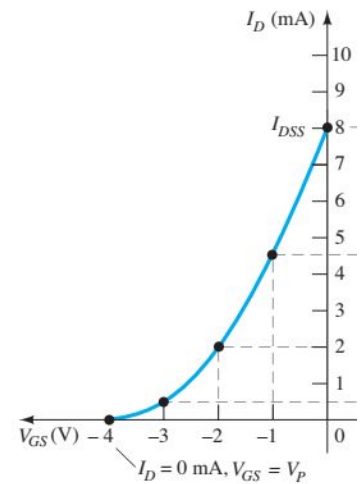
The relationship between I_D and V_{GS} is defined by *Shockley's equation*

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

control variable

constants

The squared term in the equation results in a nonlinear relationship between I_D and V_{GS} , producing a curve that grows exponentially with decreasing magnitude of V_{GS} .



Substituting $V_{GS} = 0$ V gives

$$\begin{aligned} \text{Eq. (6.3): } I_D &= I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \\ &= I_{DSS} \left(1 - \frac{0}{V_P} \right)^2 = I_{DSS} (1 - 0)^2 \end{aligned}$$

and

$$I_D = I_{DSS} |_{V_{GS}=0 \text{ V}}$$

Substituting $V_{GS} = V_P$ yields

$$\begin{aligned} I_D &= I_{DSS} \left(1 - \frac{V_P}{V_P} \right)^2 \\ &= I_{DSS} (1 - 1)^2 = I_{DSS} (0) \end{aligned}$$

$$I_D = 0 \text{ A} |_{V_{GS}=V_P}$$

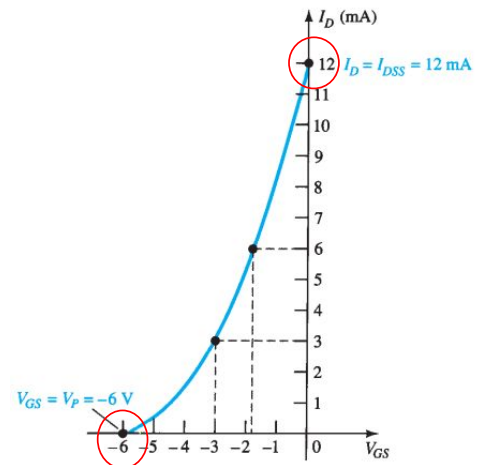


FIG. 6.18

Transfer curve for Example 6.1.

$$\begin{aligned}
 I_D &= I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \\
 &= I_{DSS} \left(\frac{1 - V_P/2}{V_P} \right)^2 = I_{DSS} \left(1 - \frac{1}{2} \right)^2 = I_{DSS} (0.5)^2 \\
 &= I_{DSS} (0.25)
 \end{aligned}$$

$$I_D = \frac{I_{DSS}}{4} \Big|_{V_{GS} = V_P/2}$$

If we choose $I_D = I_{DSS}/2$ and substitute into Eq. (6.8), we find that

$$\begin{aligned}
 V_{GS} &= V_P \left(1 - \sqrt{\frac{I_D}{I_{DSS}}} \right) \\
 &= V_P \left(1 - \sqrt{\frac{I_{DSS}/2}{I_{DSS}}} \right) = V_P (1 - \sqrt{0.5}) = V_P (0.293)
 \end{aligned}$$

and

$$V_{GS} \cong 0.3V_P \Big|_{I_D = I_{DSS}/2}$$

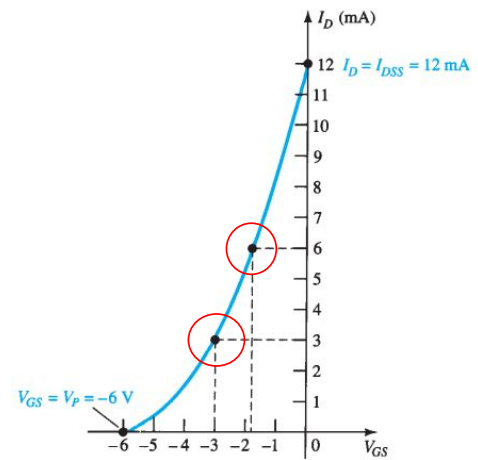


FIG. 6.18
Transfer curve for Example 6.1.

TABLE 6.1

V_{GS} versus I_D Using Shockley's Equation

V_{GS}	I_D
0	I_{DSS}
$0.3V_P$	$I_{DSS}/2$
$0.5V_P$	$I_{DSS}/4$
V_P	0 mA

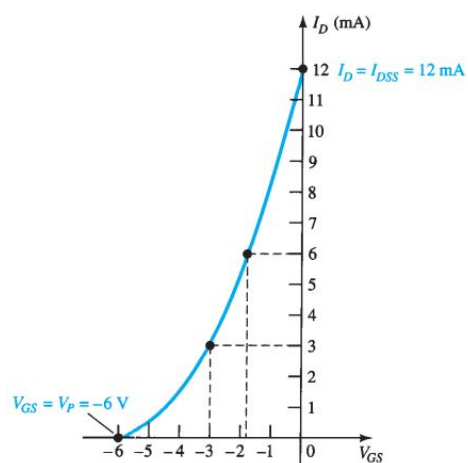


FIG. 6.18

Transfer curve for Example 6.1.

EXAMPLE 6.2 Sketch the transfer curve for a p -channel device with $I_{DSS} = 4 \text{ mA}$ and $V_P = 3 \text{ V}$.

Solution: At $V_{GS} = V_P/2 = 3 \text{ V}/2 = 1.5 \text{ V}$, $I_D = I_{DSS}/4 = 4 \text{ mA}/4 = 1 \text{ mA}$. At $I_D = I_{DSS}/2 = 4 \text{ mA}/2 = 2 \text{ mA}$, $V_{GS} = 0.3V_P = 0.3(3 \text{ V}) = 0.9 \text{ V}$. Both plot points appear in Fig. 6.19 along with the points defined by I_{DSS} and V_P .

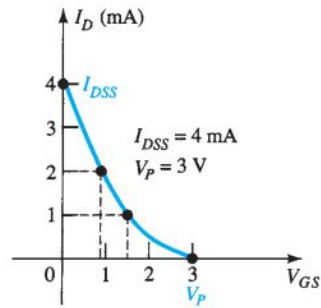


FIG. 6.19

Transfer curve for the p -channel device of Example 6.2.

DEPLETION-TYPE MOSFET

Basic Construction

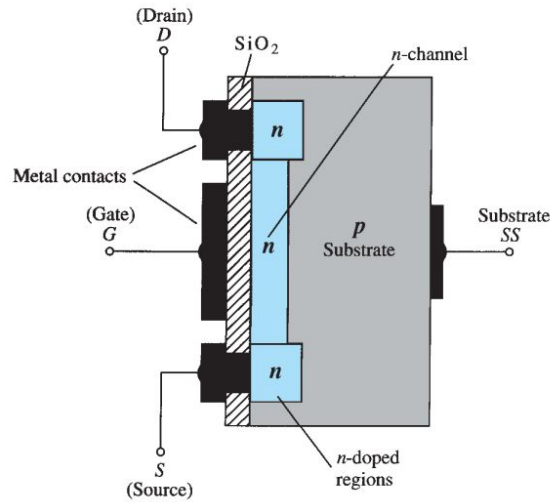


FIG. 6.24
n-Channel depletion-type MOSFET.

The basic construction of the n -channel depletion-type MOSFET is provided in Fig. 6.24. A slab of p -type material is formed from a silicon base and is referred to as the *substrate*. It is the foundation on which the device is constructed. In some cases the substrate is internally connected to the source terminal. However, many discrete devices provide an additional terminal labeled *SS*, resulting in a four-terminal device, such as that in Fig. 6.24. The source and drain terminals are connected through metallic contacts to n -doped regions linked by an n -channel as shown in the figure. The gate is also connected to a metal contact surface but remains insulated from the n -channel by a very thin silicon dioxide (SiO_2) layer. SiO_2 is a type of insulator referred to as a *dielectric*, which sets up opposing (as indicated by the prefix *di*-) electric fields within the dielectric when exposed to an externally applied field. The fact that the SiO_2 layer is an insulating layer means that:

There is no direct electrical connection between the gate terminal and the channel of a MOSFET.

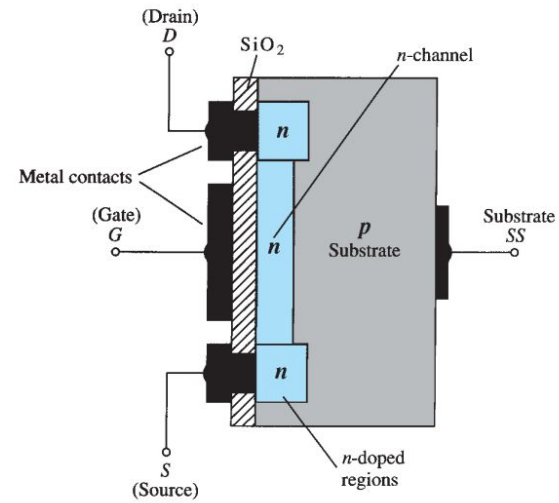


FIG. 6.24
n-Channel depletion-type MOSFET.

It is the insulating layer of SiO_2 in the MOSFET construction that accounts for the very desirable high input impedance of the device.

In fact, the input resistance of a MOSFET is usually more than that of a typical JFET, even though the input impedance of most JFETs is sufficiently high for most applications. Because of the very high input impedance, the gate current I_G is essentially 0 A for dc-biased configurations.

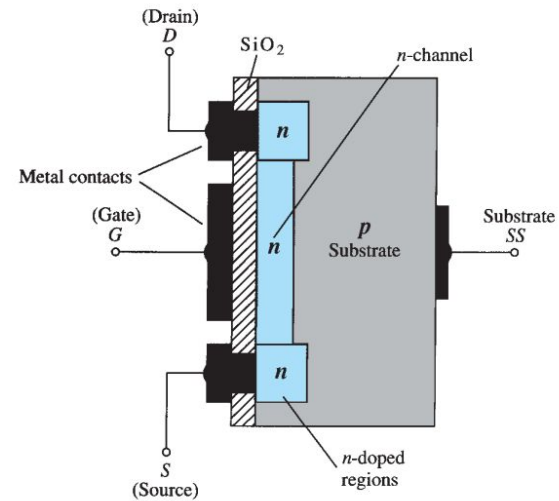


FIG. 6.24
n-Channel depletion-type MOSFET.

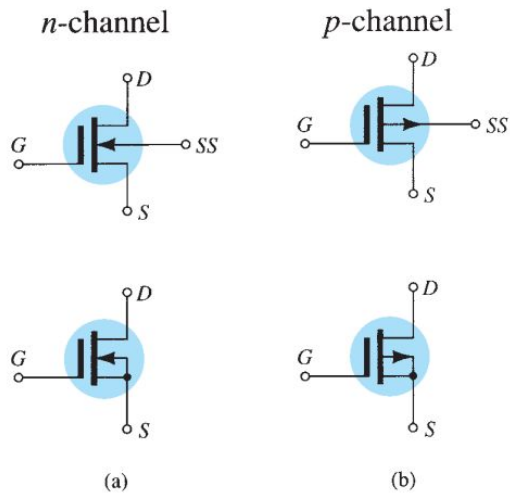


FIG. 6.30

Graphic symbols for: (a) *n*-channel depletion-type MOSFETs and (b) *p*-channel depletion-type MOSFETs.

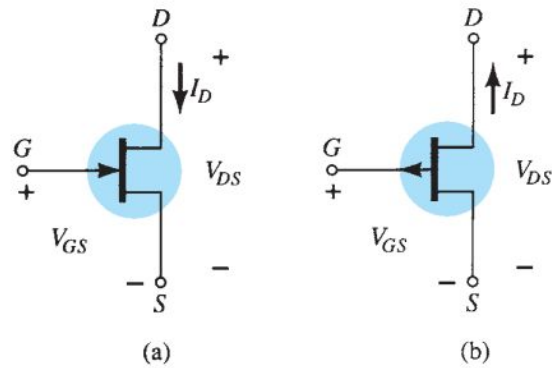


FIG. 6.14

JFET symbols: (a) *n*-channel; (b) *p*-channel.

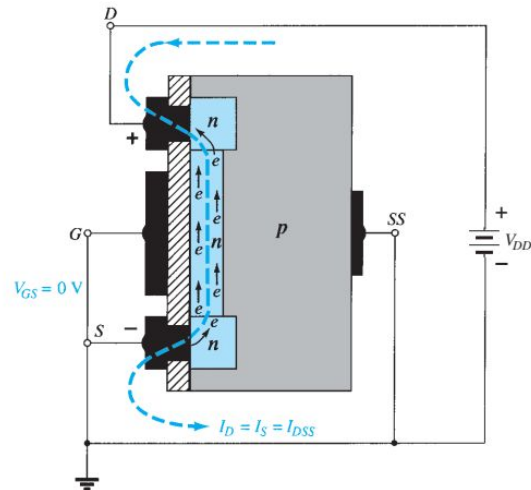


FIG. 6.25

n-Channel depletion-type MOSFET with $V_{GS} = 0$ V and applied voltage V_{DS} .

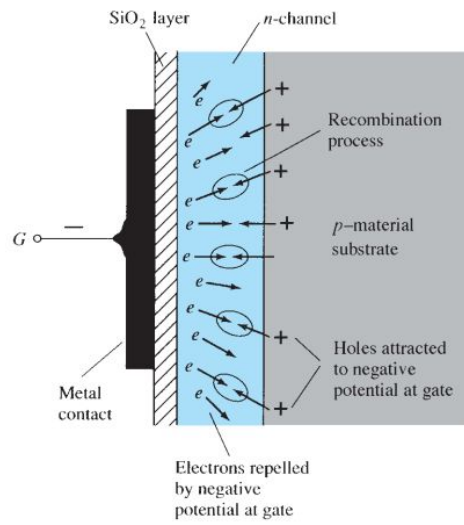


FIG. 6.27

Reduction in free carriers in a channel due to a negative potential at the gate terminal.

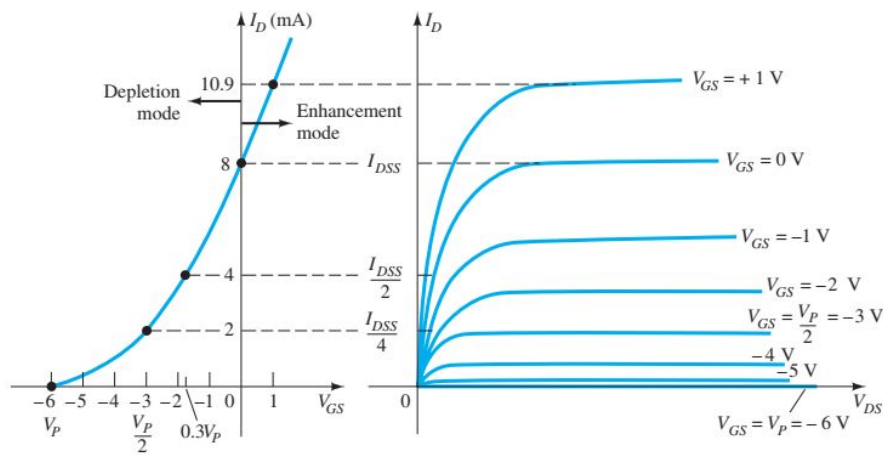


FIG. 6.26

Drain and transfer characteristics for an n-channel depletion-type MOSFET.

That is, for the device of Fig. 6.26, the application of a voltage $V_{GS} = +4$ V would result in a drain current of 22.2 mA, which could possibly exceed the maximum rating (current or power) for the device.

EXAMPLE 6.3 Sketch the transfer characteristics for an n -channel depletion-type MOSFET with $I_{DSS} = 10 \text{ mA}$ and $V_P = -4 \text{ V}$.



Solution:

At $V_{GS} = 0 \text{ V}$, $I_D = I_{DSS} = 10 \text{ mA}$

$V_{GS} = V_P = -4 \text{ V}$, $I_D = 0 \text{ mA}$

$$V_{GS} = \frac{V_P}{2} = \frac{-4 \text{ V}}{2} = -2 \text{ V}, \quad I_D = \frac{I_{DSS}}{4} = \frac{10 \text{ mA}}{4} = 2.5 \text{ mA}$$

and at $I_D = \frac{I_{DSS}}{2}$,

$$V_{GS} = 0.3V_P = 0.3(-4 \text{ V}) = -1.2 \text{ V}$$

all of which appear in Fig. 6.28.

$$\begin{aligned} I_D &= I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \\ &= (10 \text{ mA}) \left(1 - \frac{+1 \text{ V}}{-4 \text{ V}} \right)^2 = (10 \text{ mA}) (1 + 0.25)^2 = (10 \text{ mA}) (1.5625) \\ &\cong 15.63 \text{ mA} \end{aligned}$$

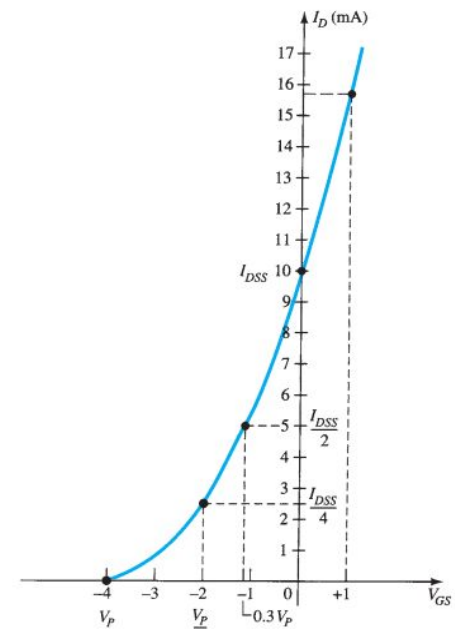


FIG. 6.28

Transfer characteristics for an n -channel depletion-type MOSFET with $I_{DSS} = 10 \text{ mA}$ and $V_P = -4 \text{ V}$.

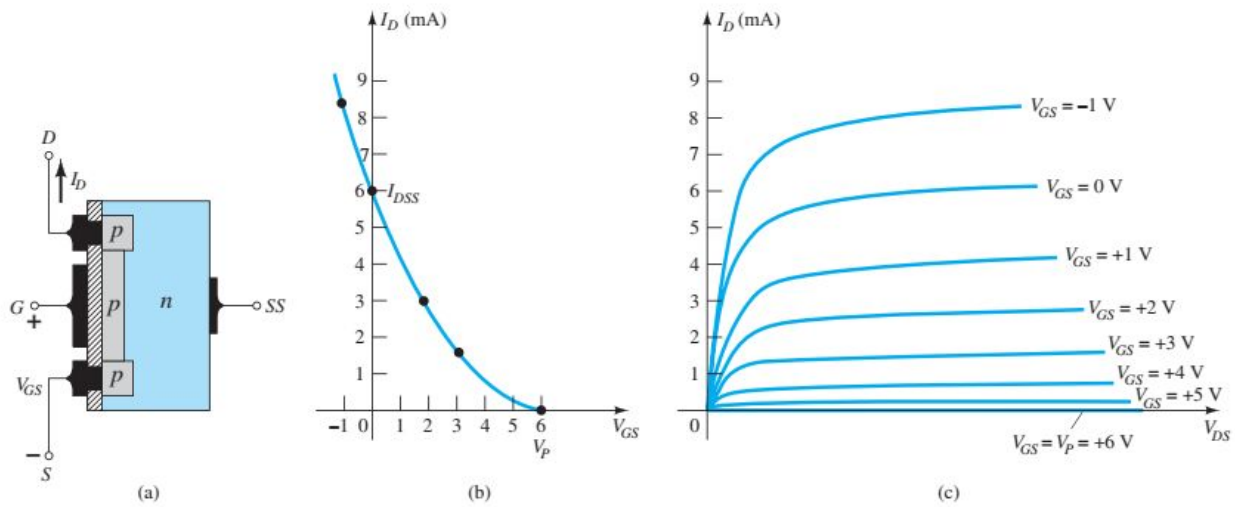


FIG. 6.29
p-Channel depletion-type MOSFET with $I_{DSS} = 6$ mA and $V_p = +6$ V.



Using an average value of 2.9 mA for the I_{DSS} of the 2N3797 MOSFET of Fig. 6.31, determine the level of V_{GS} that will result in a maximum drain current of 20 mA if $V_P = -5$ V.

$$\begin{aligned} V_{GS} &= V_P \left(1 - \sqrt{\frac{I_D}{I_{DSS}}} \right) = (-5 \text{ V}) \left(1 - \sqrt{\frac{20 \text{ mA}}{2.9 \text{ mA}}} \right) \\ &= (-5 \text{ V})(1 - 2.626) = (-5 \text{ V})(-1.626) \\ &= \mathbf{8.13 \text{ V}} \end{aligned}$$

Basic Construction

The basic construction of the n -channel enhancement-type MOSFET is provided in Fig. 6.32. A slab of p -type material is formed from a silicon base and is again referred to as the substrate. As with the depletion-type MOSFET, the substrate is sometimes internally connected to the source terminal, whereas in other cases a fourth lead (labeled SS) is made available for external control of its potential level. The source and drain terminals are again connected through metallic contacts to n -doped regions, but note in Fig. 6.32 the absence of a channel between the two n -doped regions. This is the primary difference between the construction of depletion-type and enhancement-type MOSFETs—the absence of a channel as a constructed component of the device. The SiO_2 layer is still present to isolate the gate metallic platform from the region between the drain and source, but now it is simply separated from a section of the p -type material. In summary, therefore, the construction of an enhancement-type MOSFET is quite similar to that of the depletion-type MOSFET, except for the absence of a channel between the drain and source terminals.

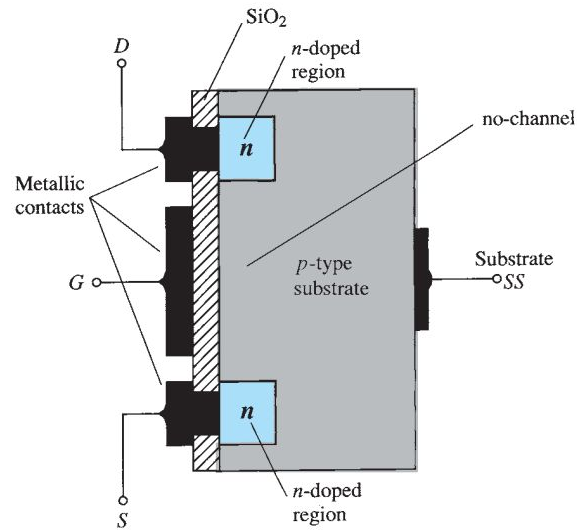


FIG. 6.32

n-Channel enhancement-type MOSFET.

Basic Operation and Characteristics

If V_{GS} is set at 0 V and a voltage applied between the drain and the source of the device of Fig. 6.32, the absence of an n -channel (with its generous number of free carriers) will result in a current of effectively 0 A—quite different from the depletion-type MOSFET and JFET,

With V_{DS} some positive voltage, V_{GS} at 0 V, and terminal SS directly connected to the source, there are in fact two reverse-biased p - n junctions between the n -doped regions and the p -substrate to oppose any significant flow between drain and source.

In Fig. 6.33, both V_{DS} and V_{GS} have been set at some positive voltage greater than 0 V, establishing the drain and the gate at a positive potential with respect to the source.

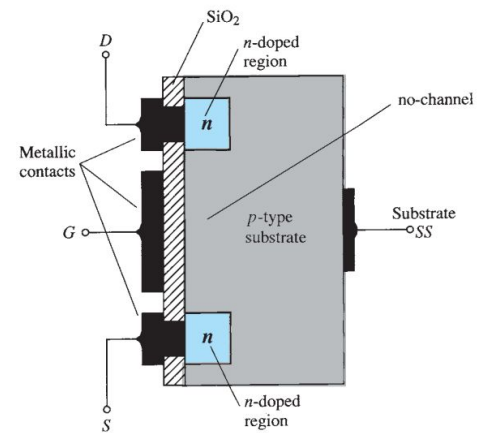


FIG. 6.32

n-Channel enhancement-type MOSFET.

Basic Operation and Characteristics

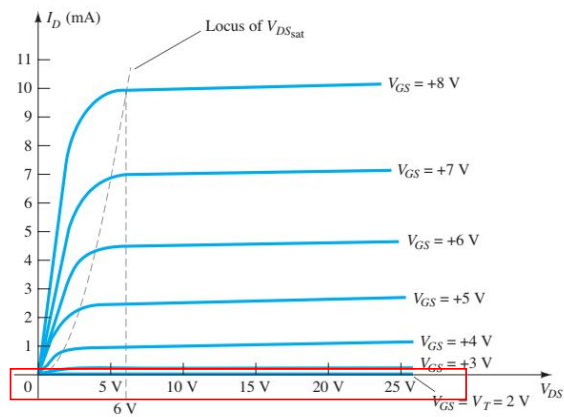


FIG. 6.35

Drain characteristics of an n-channel enhancement-type MOSFET with $V_T = 2$ V and $k = 0.278 \times 10^{-3}$ A/V².

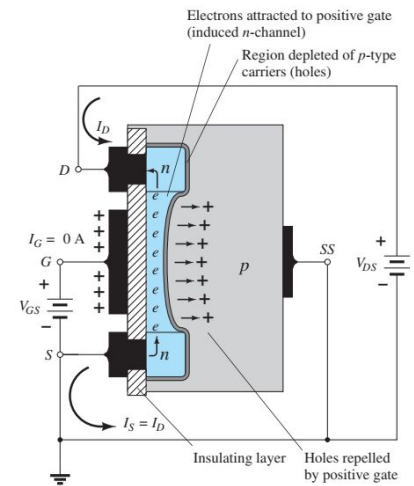


FIG. 6.33

Channel formation in the n-channel enhancement-type MOSFET.

Basic Operation and Characteristics

The positive potential at the gate will pressure the holes (since like charges repel) in the p -substrate along the edge of the SiO_2 layer to leave the area and enter deeper regions of the p -substrate, as shown in the figure. The result is a depletion region near the SiO_2 insulating layer void of holes. However, the electrons in the p -substrate (the minority carriers of the material) will be attracted to the positive gate and accumulate in the region near the surface of the SiO_2 layer. **The SiO_2 layer and its insulating qualities will prevent the negative carriers from being absorbed at the gate terminal.** As V_{GS} increases in magnitude, the concentration of electrons near the SiO_2 surface increases until eventually the induced n -type region can support a measurable flow between drain and source. The level of V_{GS} that results in the significant increase in drain current is called the *threshold voltage* and is given the symbol V_T . On specification sheets it is referred to as $V_{GS(\text{Th})}$, although V_T is less unwieldy and will be used in the analysis to follow. Since the channel is nonexistent with $V_{GS} = 0$ V and “enhanced” by the application of a positive gate-to-source voltage, this type of MOSFET is called an *enhancement-type MOSFET*. Both depletion- and enhancement-type MOSFETs have enhancement-type regions, but the label was applied to the latter since it is its only mode of operation.

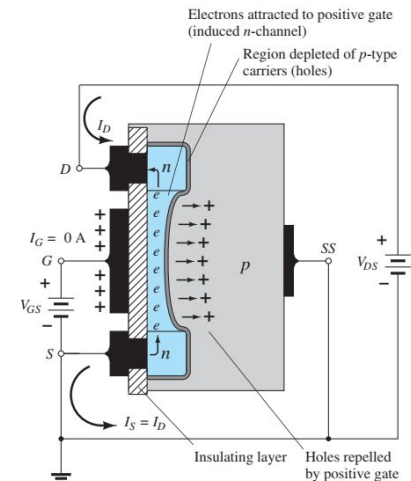
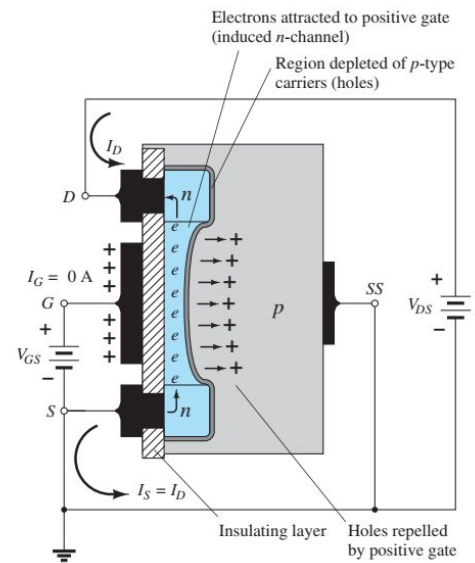
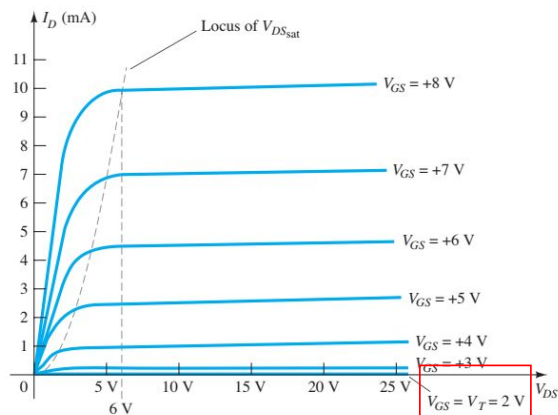


FIG. 6.33

Channel formation in the n -channel enhancement-type MOSFET.

Basic Operation and Characteristics



Basic Operation and Characteristics

As V_{GS} is increased beyond the threshold level, the density of free carriers in the induced channel will increase, resulting in an increased level of drain current. However, if we hold V_{GS} constant and increase the level of V_{DS} , the drain current will eventually reach a saturation level as occurred for the JFET and depletion-type MOSFET. The leveling off of I_D is due to a pinching-off process depicted by the narrower channel at the drain end of the induced channel as shown in Fig. 6.34. Applying Kirchhoff's voltage law to the terminal voltages of the MOSFET of Fig. 6.34, we find that

$$V_{DG} = V_{DS} - V_{GS} \quad (6.13)$$

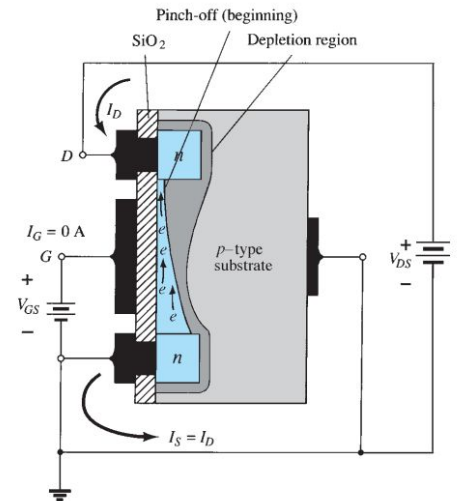


FIG. 6.34

Change in channel and depletion region with increasing level of V_{DS} for a fixed value of V_{GS} .

Basic Operation and Characteristics

If V_{GS} is held fixed at some value such as 8 V and V_{DS} is increased from 2 V to 5 V, the voltage V_{DG} [by Eq. (6.13)] will increase from -6 V to -3 V and the gate will become less and less positive with respect to the drain. This reduction in gate-to-drain voltage will in turn reduce the attractive forces for free carriers (electrons) in this region of the induced channel, causing a reduction in the effective channel width. Eventually, the channel will be reduced to the point of pinch-off and a saturation condition will be established as described earlier for the JFET and depletion-type MOSFET. In other words, any further increase in V_{DS} at the fixed value of V_{GS} will not affect the saturation level of I_D until breakdown conditions are encountered.

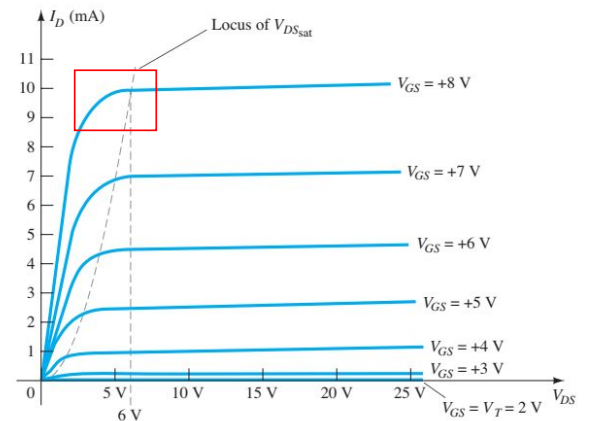


FIG. 6.35

Drain characteristics of an n-channel enhancement-type MOSFET with $V_T = 2 \text{ V}$ and $k = 0.278 \times 10^{-3} \text{ A/V}^2$.

Transfer Characteristics Curve

For levels of $V_{GS} > V_T$, the drain current is related to the applied gate-to-source voltage by the following nonlinear relationship:

$$I_D = k(V_{GS} - V_T)^2 \quad (6.15)$$

Again, it is the squared term that results in the nonlinear (curved) relationship between I_D and V_{GS} . The k term is a constant that is a function of the construction of the device. The value of k can be determined from the following equation [derived from Eq. (6.15)], where $I_{D(on)}$ and $V_{GS(on)}$ are the values for each at a particular point on the characteristics of the device.

$$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_T)^2} \quad (6.16)$$

Transfer Characteristics Curve

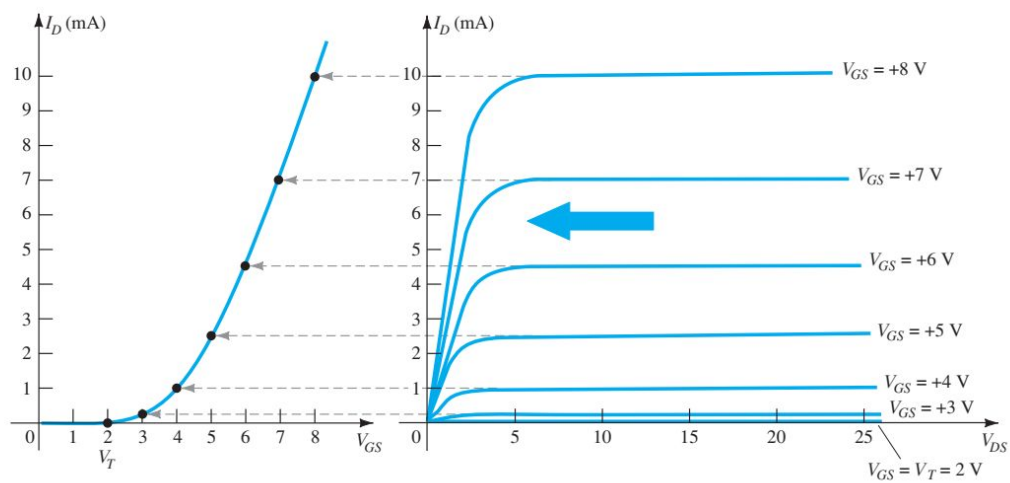


FIG. 6.36

Sketching the transfer characteristics for an n-channel enhancement-type MOSFET from the drain characteristics.

Transfer Characteristics Curve

EXAMPLE 6.4 Using the data provided on the specification sheet of Fig. 6.40 and an average threshold voltage of $V_{GS(Th)} = 3 \text{ V}$, determine:

- The resulting value of k for the MOSFET.
- The transfer characteristics.

Solution:

a. Eq. (6.16):
$$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_{GS(Th)})^2}$$
$$= \frac{3 \text{ mA}}{(10 \text{ V} - 3 \text{ V})^2} = \frac{3 \text{ mA}}{(7 \text{ V})^2} = \frac{3 \times 10^{-3}}{49} \text{ A/V}^2$$
$$= 0.061 \times 10^{-3} \text{ A/V}^2$$

Transfer Characteristics Curve

b. Eq. (6.15):

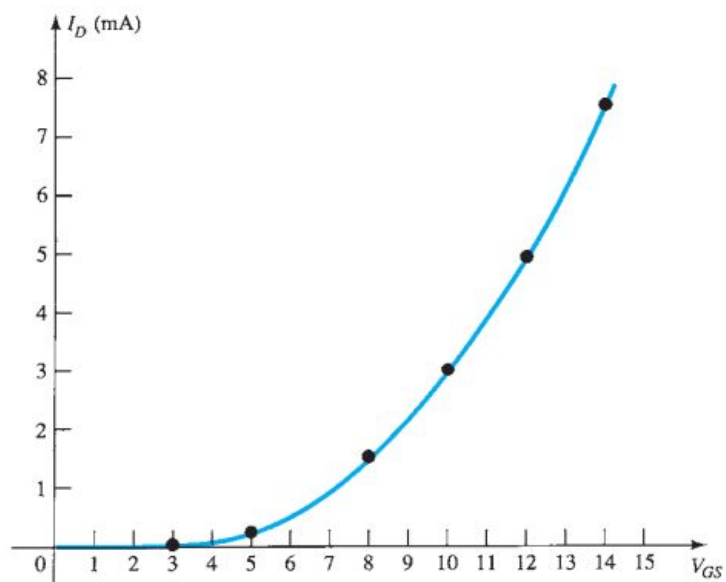
$$I_D = k(V_{GS} - V_T)^2$$
$$= 0.061 \times 10^{-3}(V_{GS} - 3 \text{ V})^2$$

For $V_{GS} = 5 \text{ V}$,

$$I_D = 0.061 \times 10^{-3}(5 \text{ V} - 3 \text{ V})^2 = 0.061 \times 10^{-3}(2)^2$$
$$= 0.061 \times 10^{-3}(4) = 0.244 \text{ mA}$$

For $V_{GS} = 8, 10, 12$, and 14 V , I_D will be 1.525, 3 (as defined), 4.94, and 7.38 mA, respectively. The transfer characteristics are sketched in Fig. 6.41.

Transfer Characteristics Curve





Assignment (15 Marks)

CMOS

FET Biasing



The general relationships that can be applied to the dc analysis of all FET amplifiers are

$$I_G \cong 0 \text{ A} \quad (7.1)$$

and

$$I_D = I_S \quad (7.2)$$

For JFETs and depletion-type MOSFETs and MESFETs, Shockley's equation is applied to relate the input and output quantities:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \quad (7.3)$$

For enhancement-type MOSFETs and MESFETs, the following equation is applicable:

$$I_D = k(V_{GS} - V_T)^2 \quad (7.4)$$

7.2 FIXED-BIAS CONFIGURATION

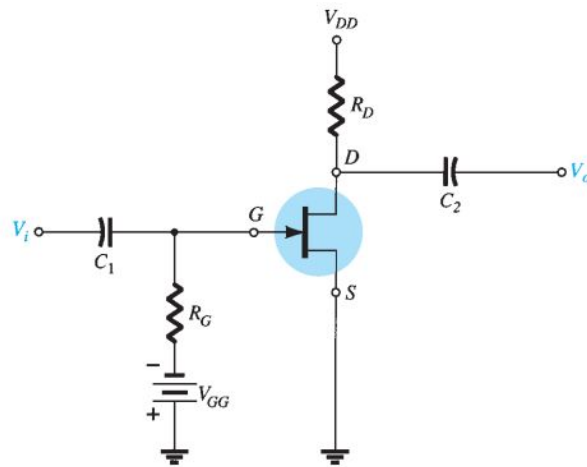


FIG. 7.1

Fixed-bias configuration.

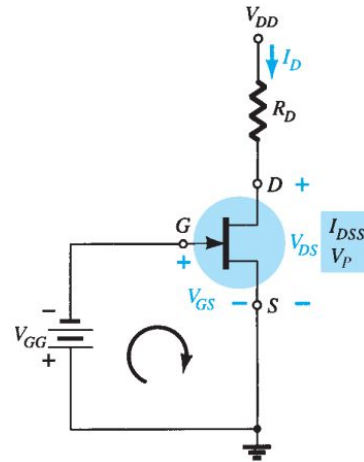


FIG. 7.2

Network for dc analysis.

7.2 FIXED-BIAS CONFIGURATION

$$I_G \cong 0 \text{ A}$$

$$V_{R_G} = I_G R_G = (0 \text{ A}) R_G = 0 \text{ V}$$

$$-V_{GG} - V_{GS} = 0$$

$$V_{GS} = -V_{GG}$$

$$+V_{DS} + I_D R_D - V_{DD} = 0$$

$$V_{DS} = V_{DD} - I_D R_D$$

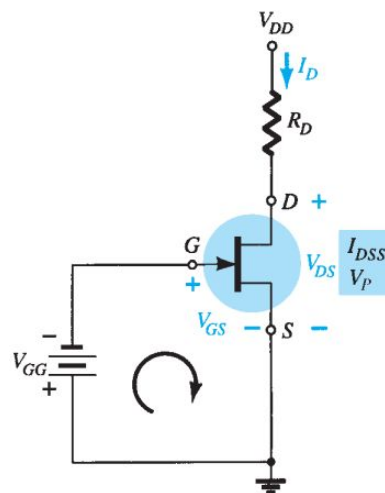


FIG. 7.2

Network for dc analysis.

7.2 FIXED-BIAS CONFIGURATION

$$V_S = 0 \text{ V}$$

$$V_{DS} = V_D - V_S$$

$$V_D = V_{DS} + V_S = V_{DS} + 0 \text{ V}$$

$$V_D = V_{DS}$$

$$V_{GS} = V_G - V_S$$

$$V_G = V_{GS} + V_S = V_{GS} + 0 \text{ V}$$

$$V_G = V_{GS}$$

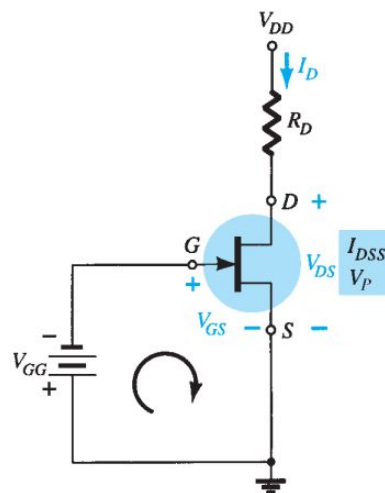


FIG. 7.2
Network for dc analysis.

7.2 FIXED-BIAS CONFIGURATION

EXAMPLE 7.1 Determine the following for the network of Fig. 7.6:

- V_{GS_Q} .
- I_{D_Q} .
- V_{DS} .
- V_D .
- V_G .
- V_S .

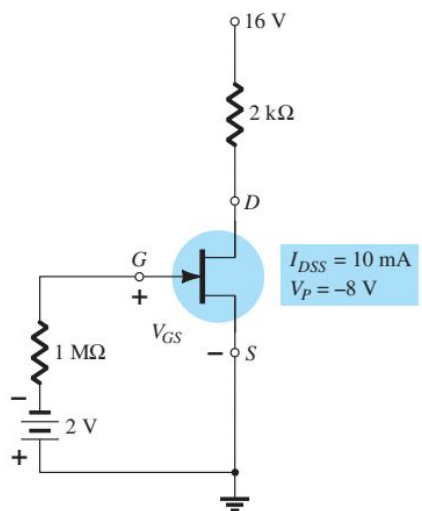


FIG. 7.6

Example 7.1.



7.2 FIXED-BIAS CONFIGURATION

Solution:

Mathematical Approach

a. $V_{GS_Q} = -V_{GG} = -2 \text{ V}$

b. $I_{D_Q} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 = 10 \text{ mA} \left(1 - \frac{-2 \text{ V}}{-8 \text{ V}} \right)^2$
 $= 10 \text{ mA} (1 - 0.25)^2 = 10 \text{ mA} (0.75)^2 = 10 \text{ mA} (0.5625)$
 $= \mathbf{5.625 \text{ mA}}$

c. $V_{DS} = V_{DD} - I_D R_D = 16 \text{ V} - (5.625 \text{ mA})(2 \text{ k}\Omega)$
 $= 16 \text{ V} - 11.25 \text{ V} = \mathbf{4.75 \text{ V}}$

d. $V_D = V_{DS} = \mathbf{4.75 \text{ V}}$

e. $V_G = V_{GS} = -2 \text{ V}$

f. $V_S = \mathbf{0 \text{ V}}$

7.2 FIXED-BIAS CONFIGURATION

Graphical Approach The resulting Shockley curve and the vertical line at $V_{GS} = -2\text{ V}$ are provided in Fig. 7.7. It is certainly difficult to read beyond the second place without

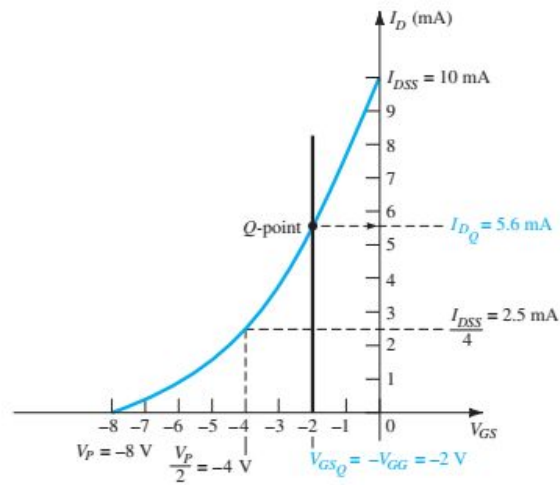


FIG. 7.7

Graphical solution for the network of Fig. 7.6.

7.3 SELF-BIAS CONFIGURATION

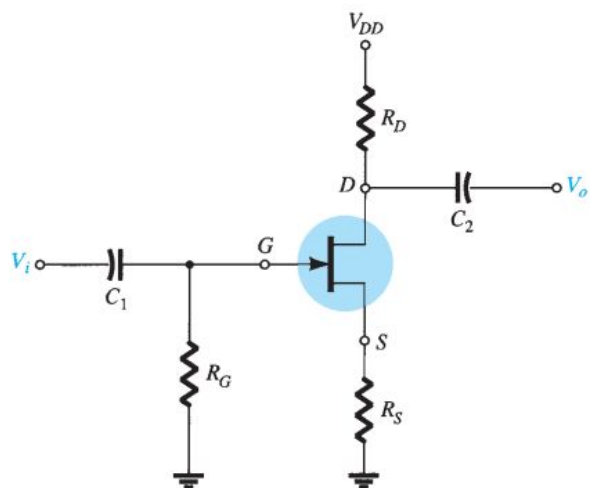


FIG. 7.8
JFET self-bias configuration.

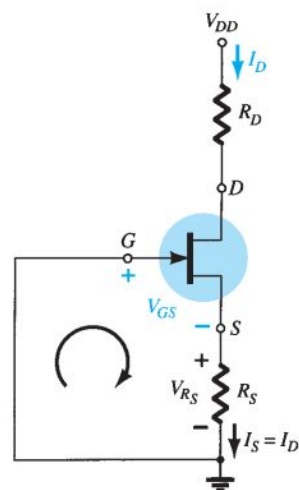


FIG. 7.9
DC analysis of the self-bias configuration.

7.3 SELF-BIAS CONFIGURATION

The current through R_S is the source current I_S , but $I_S = I_D$ and

$$V_{R_S} = I_D R_S$$

For the indicated closed loop of Fig. 7.9, we find that

$$-V_{GS} - V_{R_S} = 0$$

and

$$V_{GS} = -V_{R_S}$$

or

$$V_{GS} = -I_D R_S$$

$$I_D = I_S$$

$$V_{DS} = V_{DD} - I_D (R_S + R_D)$$

$$V_S = I_D R_S$$

$$V_G = 0 \text{ V}$$

$$V_D = V_{DS} + V_S = V_{DD} - V_{R_D}$$

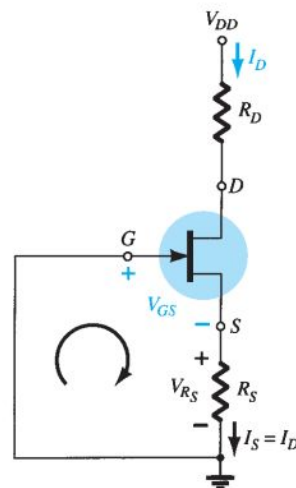


FIG. 7.9

DC analysis of the self-bias configuration.

7.3 SELF-BIAS CONFIGURATION



EXAMPLE 7.2 Determine the following for the network of Fig. 7.12:

- V_{GS_Q}
- I_{D_Q}
- V_{DS}
- V_S
- V_G
- V_D

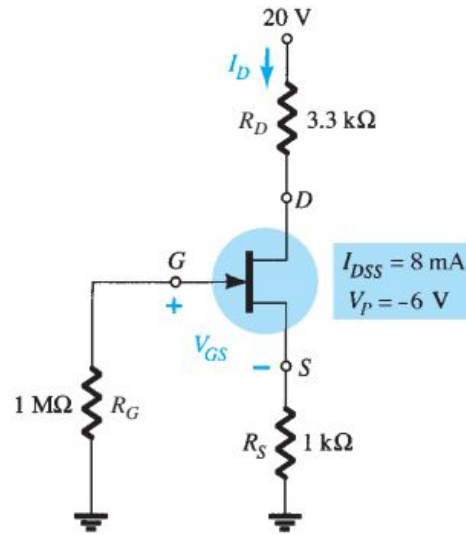


FIG. 7.12
Example 7.2.

7.3 SELF-BIAS CONFIGURATION



Solution:

- a. The gate-to-source voltage is determined by

$$V_{GS} = -I_D R_S$$

Choosing $I_D = 4 \text{ mA}$, we obtain

$$V_{GS} = -(4 \text{ mA})(1 \text{ k}\Omega) = -4 \text{ V}$$

The result is the plot of Fig. 7.13 as defined by the network.

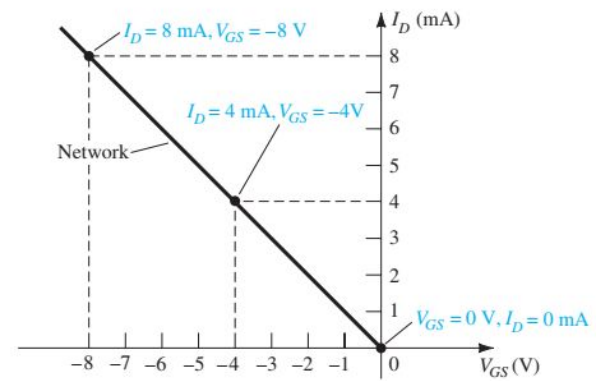


FIG. 7.13

Sketching the self-bias line for the network of Fig. 7.12.

7.3 SELF-BIAS CONFIGURATION

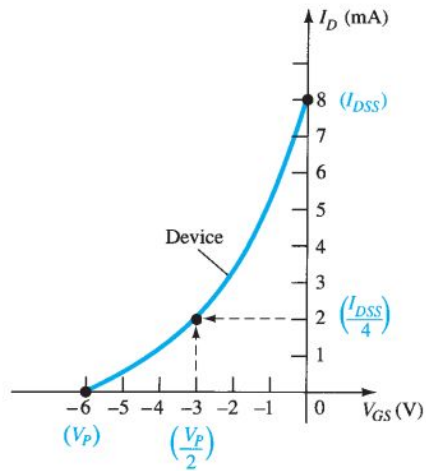


FIG. 7.14

Sketching the device characteristics for the JFET of Fig. 7.12.

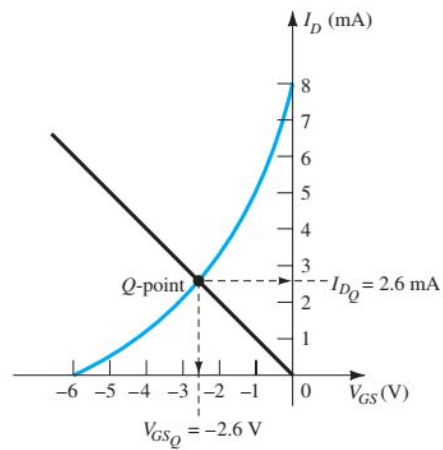


FIG. 7.15

Determining the Q -point for the network of Fig. 7.12.

7.3 SELF-BIAS CONFIGURATION



b. At the quiescent point

$$I_{D_Q} = 2.6 \text{ mA}$$

c. Eq. (7.11): $V_{DS} = V_{DD} - I_D(R_S + R_D)$
 $= 20 \text{ V} - (2.6 \text{ mA})(1 \text{ k}\Omega + 3.3 \text{ k}\Omega)$
 $= 20 \text{ V} - 11.18 \text{ V}$
 $= 8.82 \text{ V}$

d. Eq. (7.12): $V_S = I_D R_S$
 $= (2.6 \text{ mA})(1 \text{ k}\Omega)$
 $= 2.6 \text{ V}$

e. Eq. (7.13): $V_G = 0 \text{ V}$

f. Eq. (7.14): $V_D = V_{DS} + V_S = 8.82 \text{ V} + 2.6 \text{ V} = 11.42 \text{ V}$
or $V_D = V_{DD} - I_D R_D = 20 \text{ V} - (2.6 \text{ mA})(3.3 \text{ k}\Omega) = 11.42 \text{ V}$

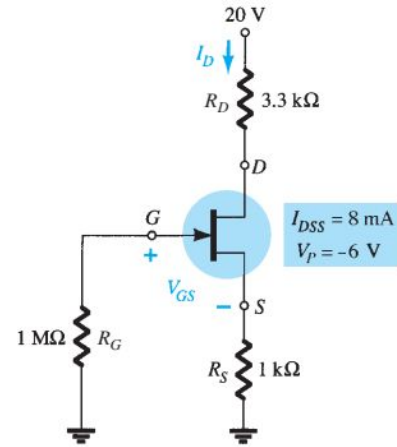


FIG. 7.12

Example 7.2.

7.3 SELF-BIAS CONFIGURATION



EXAMPLE 7.3 Find the quiescent point for the network of Fig. 7.12 if:

- $R_S = 100 \Omega$.
- $R_S = 10 \text{ k}\Omega$.

Solution: Both $R_S = 100 \Omega$ and $R_S = 10 \text{ k}\Omega$ are plotted on Fig. 7.16.

- For $R_S = 100 \Omega$:

$$I_{D_Q} \cong 6.4 \text{ mA}$$

and from Eq. (7.10),

$$V_{GS_Q} \cong -0.64 \text{ V}$$

- For $R_S = 10 \text{ k}\Omega$

$$V_{GS_Q} \cong -4.6 \text{ V}$$

and from Eq. (7.10),

$$I_{D_Q} \cong 0.46 \text{ mA}$$

In particular, note how lower levels of R_S bring the load line of the network closer to the I_D axis, whereas increasing levels of R_S bring the load line closer to the V_{GS} axis.

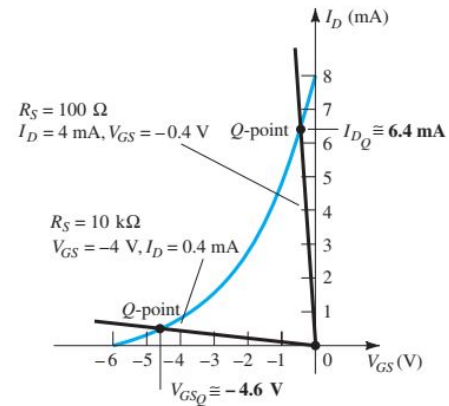


FIG. 7.16
Example 7.3.